

FEDERAL INSTITUTE OF SCIENCE AND TECHNOLOGY (FISAT)
DEPARTMENT OF COMPUTER APPLICATIONS
MAIN PROJECT
SCRUM BOOK

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Project Title: FISAT GATEWAY MANAGEMENT SYSTEM

GitHub ID: <https://github.com/shafnashakeersm/Main-Project>

Sprint Release #: 1**Date: 20/02/2025****Description of Work:**

The **FISAT Gateway Management System** is a web-based application designed to digitize and streamline the manual processes involved in managing visitor records, hostel student movements, and staff gate transactions within FISAT College. Built using the MERN stack (MongoDB, Express.js, React.js, Node.js), the system ensures efficient tracking, secure record-keeping, and seamless communication among stakeholders, including wardens, watchmen, parents, and administrators. The platform allows wardens to add and manage student details, including personal information and leave requests, which require parental approval via API-based notifications before students can exit the campus. Once a parent approves, the leave request is forwarded to watchman for final clearance. Additionally, the system enables students traveling by train to upload their ticket details along with their expected return time. The watchman interface provides a real-time view of approved student movements and visitor logs, ensuring that only authorized individuals enter or leave the premises. The admin dashboard consolidates daily reports, including visitor logs, student leave records, and watchman reports, providing complete visibility over campus movements. By implementing role-based access control, the system ensures that wardens, watchmen, and administrators have functionalities tailored to their responsibilities. The integration of automated notifications and approvals enhances transparency, reduces manual errors, and improves overall security within the institution.

Remarks:**Name and Signature of the Guide****Name and Signature of the Scrum Master**

Date: 20/02/2025

PRODUCT BACKLOG

>> User Authentication & Role Management

- Implemented a secure login system for different user roles, including Admin, Wardens, and Watchmen.
- Ensured role-based access control so that each user only interacts with the necessary functionalities.

>> Student Management Module

- Wardens can add and manage student records, including student ID, name, batch, class, and parental contact information.
- Allows fetching of student details by ID for quick access.

>> Student Leave & Approval Module

- When a student requests leave, the system notifies parents via API-based messages.
- Parents can approve or reject the request directly through a mobile notification.
- Once approved, the warden forwards the request to the watchman.
- If students are traveling by train, they can upload ticket details along with expected return time.

>> Watchman Dashboard & Visitor Management

- The watchman receives a real-time list of approved students who are allowed to exit.
- Stores visitor details, including name, purpose of visit, entry and exit time for security tracking.

>> Notification & Messaging System

- Automated API-based notifications are sent to parents when students leave the hostel.
- Watchmen receive updates regarding student movements.
- Ensures all approvals and actions are logged for future reference.

>> Admin Dashboard & Report Generation

- The admin can view reports on daily visitor logs, student leave details, and watchman reports.
- Provides an overview of student movements, ensuring efficient monitoring and record-keeping.

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DATABASE & UI DESIGN**DATABASE****>> Student Details**

Field Name	Data Type	Description
<code>_id</code>	ObjectId	Unique ID (Auto-generated by MongoDB)
<code>studentID</code>	String	Unique Student ID
<code>name</code>	String	Student's Full Name
<code>batch</code>	String	Student's Batch (e.g., 2023)
<code>class</code>	String	Class/Department (e.g., CSE, ECE)
<code>parentPhone</code>	String	Parent's Contact Number
<code>createdAt</code>	Date	Record Creation Timestamp

>> Student Leave Record

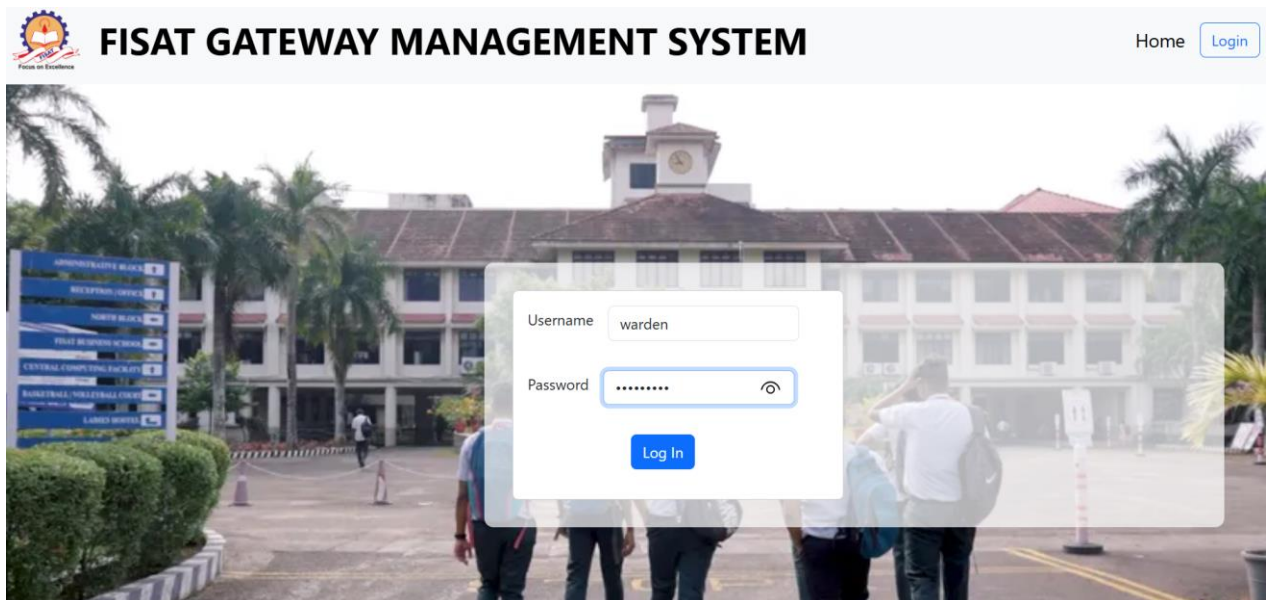
Field Name	Data Type	Description
<code>_id</code>	ObjectId	Unique ID (Auto-generated)
<code>studentID</code>	String	Student's Unique ID (Reference to <code>students</code> collection)
<code>studentName</code>	String	Student's Name
<code>batch</code>	String	Student's Batch
<code>class</code>	String	Student's Class
<code>parentApproval</code>	Boolean	Whether Parent Approved (true/false)
<code>approvalStatus</code>	String	Status: Pending, Approved, or Rejected
<code>departureTime</code>	Date	Time Student Leaves the Campus
<code>returnTime</code>	Date	Expected Return Time
<code>ticketUploaded</code>	Boolean	If the Train Ticket is Uploaded
<code>createdAt</code>	Date	Record Creation Timestamp

>> Watchman Report

Field Name	Data Type	Description
<code>_id</code>	ObjectId	Unique ID
<code>date</code>	Date	Report Date
<code>studentMovements</code>	Array of Objects	List of students who left/returned
Inside <code>studentMovements</code> Array:		
<code>studentID</code>	String	Student ID (Reference to <code>students</code> collection)
<code>name</code>	String	Student's Name
<code>departureTime</code>	Date	Time of Exit
<code>returnTime</code>	Date	Time of Return

>> Login and Role Management

Field Name	Data Type	Description
<code>_id</code>	ObjectId	Unique ID
<code>username</code>	String	User's Username (Unique)
<code>password</code>	String	Encrypted Password (Hashed)
<code>role</code>	String	Role: Admin, Warden, or Watchman
<code>createdAt</code>	Date	Account Creation Date

UI DESIGN**Men's Warden Dashboard****Fetch Student Details**

Enter Student ID
Fetch Student Details

Manage Students**Add Student**

Name
Student ID
Batch
Class
Parent's Phone Number
Add Student

List of Added Students

- **John Doe** (ID: STU12345) - A, Class: CSE S3, Parent Phone: 9876543210
- **Maya Dev** (ID: STU12876) - B, Class: MCA S2, Parent Phone: 7865436789

Name and Signature of the Guide:

Name and Signature of the Scrum Master

Date:

TESTING AND VALIDATION

Name and Signature of the Guide:

Name and Signature of the Scrum Master

Date:

DETAILS OF VERSIONS

Name and Signature of the Guide

Name and Signature of the Scrum Master